Exp. no : 06
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Hamming code - error detection and correction.

Aim:

To write hamming code for error detection and correction

Code:

```
def hamming_encryption(data):
    k = len(data)
    r = 0
    while (2**r) < (k + r + 1):
        r += 1
    r -= 1
    n = k + r + 1
    code = [0]*n
    j = 0
    for i in range(1, n):
        if (i & (i-1)) != 0:
            code[i] = data[j]
            j += 1
    for i in range(r):
        pos = 2**i
        pair = 0
        for j in range(1, n):
            if j & pos:
                pair = pair ^ int(code[j])
        code[pos] = str(pair)
    return (code[1:])
```

O/P

```
def hamming code - decryption
                (data, r, k):

    for i in range(r):
        pos = 2**i:
        pair = 0
        for j in range (1, k+k+1):
            if j & pos:
                pair ^= int (code [j])
        if (pair != 0):
            if (data [pos] == 0):
                data [pos] = '1'
            else:
                data [pos] = '0'

    return ''.join (data)
    data = input ("Enter binary
                    data:")
print ("Data after encoding
        from sender side:",
        hamming - encryption)
rdata = input ('Enter data
                received by the
                receiver")

print (hamming code - decryption
        (rdata, r, rdatalength))
```

O/P :

Enter binary data = 1011
Data after encoding from senders side = 0110 0110
Enter data received by receiver: 0.110111
Data after decoding : 0110011

Result :

Thus, the hamming code was detected and connected the error successfully.